

Responsible & Trustworthy AI — Learning Resources

Introduction

Hi everyone,

I'm **Aashish Pokharel**, and I'll be your instructor for the module **Responsible and Trustworthy AI**. I work as an ML Engineer at Fusemachines, where I build and deploy machine learning systems. Through my work, I've learned that having a strong understanding of the fundamentals makes it much easier to solve real-world machine learning problems — and that a model which merely *works* is not the same as one that can be **trusted**.

We'll start with the foundations: what it actually means for a system to be trustworthy — fairness, transparency, accountability, robustness, privacy and reliability — and the risk frameworks, the **NIST AI RMF** and the **EU AI Act**, that turn those principles into obligations. From there we'll open up the black box with **Explainable AI** (feature importance, SHAP/LIME, partial-dependence plots and counterfactuals), separate correlation from cause with **Causal AI**, and finish with the engineering and governance that keep a live system safe — input and output guardrails, least-privilege permissions, human-in-the-loop control, monitoring and tamper-evident audit.

Throughout, my goal is to help you build an intuitive understanding of these ideas, not just show you how to implement them. We'll pair every concept with runnable code and keep returning to a single running case, so the ideas have somewhere concrete to land. By the end, you'll be able to look at a real system and argue — with evidence, not intuition — whether it should ship, and who owns the risk if it does. I'm excited to have you in the course, and let's get started!

Syllabus

The module runs over four sessions, each pairing a concept with runnable code, with a hands-on engineering track running alongside.

- **Session 1 — Foundations of Responsible & Trustworthy AI.** Core principles (fairness, transparency, accountability, robustness, privacy, reliability); trustworthiness vs. accuracy; the **NIST AI RMF** and the **EU AI Act / GDPR**; the AI lifecycle.
- **Session 2 — Explainable AI (XAI).** Interpretable vs. black-box models; LIME, SHAP, partial-dependence plots and counterfactuals; global vs. local explanations; what makes an explanation good.
- **Session 3 — Causal AI for Deeper Trust** (*optional*). Correlation vs. causation; the ladder of causation; DAGs and identification; causal failure modes.
- **Session 4 — Integration, Auditing & Regulation.** Accountability as a mechanism; the evidence pack; post-market monitoring; incident response; go/no-go and residual-risk acceptance; red-teaming.
- **Engineering track — Prompt & Agentic AI Safety, plus Testing & Best Practices.** Input/output guardrails, prompt-injection defence, PII and sensitive-data handling, least-privilege permissions and human-in-the-loop control; test strategy, monitoring & observability, deployment and the best-practice checklist.

Questions we will try to answer

The module is organised around four questions. Each is anchored in one session but **cannot be answered inside it alone** — a strong answer reaches across the others. By the end, you should be able to move between them without being told that they connect.

- **What licenses the claim that a system is trustworthy?** What evidence — and whose authority — lets us call a system trustworthy at all?
- **What makes an explanation adequate — and to whom?** Why a *plausible* explanation is not the same as a *true* one, and how "adequate" changes with the audience.

- **When may a system act on a number?** What separates a correlation a model has learned from a cause it can safely act on.
- **Who is accountable once the system is live, and by what mechanism?** How oversight, audit and escalation turn accountability from a promise into a mechanism.

Learning Objectives

By the end of this module, all learners will be able to:

1. **Explain the core principles** of responsible and trustworthy AI — fairness, transparency, accountability, robustness/safety, privacy and reliability — and locate them in real deployments governed by the **NIST AI RMF** and the **EU AI Act**. (Bloom's: Remember/Understand)
2. **Apply explainability techniques** (feature importance, SHAP/LIME, partial-dependence plots, counterfactuals) to interpret a pre-trained model's predictions and state the strengths and limits of each. (Bloom's: Apply/Analyze)
3. **Distinguish correlation from causation** using the ladder of causation (association -> intervention -> counterfactual), read a Directed Acyclic Graph (DAG), and identify where causal reasoning makes a system more trustworthy. (Bloom's: Understand/Analyze)
4. **Implement input and output guardrails** — PII detection/redaction, prompt-injection detection, sensitive-topic classification, and schema plus groundedness validation — and explain why each layer exists. (Bloom's: Apply)
5. **Enforce least-privilege tool permissions and human-in-the-loop control** in an agentic system, and estimate the blast radius of a runaway agent. (Bloom's: Apply/Evaluate)
6. **Instrument a trustworthy pipeline** with tamper-evident audit logging, monitoring, incident review and red-teaming, and prove its behaviour with automated tests. (Bloom's: Apply/Evaluate)

By the end of the module, some learners will be able to (depth / stretch):

7. **Compare multiple explanation methods** (e.g., SHAP vs. counterfactuals) and evaluate them with fidelity/stability metrics or a small human-subject study. (Bloom's: Analyze/Evaluate/Create)
 8. **Build and query a simple causal model** (DAG plus estimand, DoWhy-style) for intervention or counterfactual analysis on a dataset. (Bloom's: Apply/Create)
 9. **Critique a live system** against the NIST AI RMF and the EU AI Act, assemble an evidence pack, and defend a go/no-go deployment decision with named owners, tolerances and review dates. (Bloom's: Evaluate/Create)
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Contents to cover

Topic 1 — Foundations of Responsible & Trustworthy AI

- **Core principles of trustworthy AI**
 - Fairness (bias detection and mitigation)
 - Transparency
 - Accountability
 - Robustness / safety
 - Privacy (e.g., differential privacy)
 - Reliability
- **Trustworthiness vs. accuracy** — why a highly accurate model can still be untrustworthy
- **Risk management with the NIST AI RMF** — GOVERN -> MAP -> MEASURE -> MANAGE

- **Regulatory frame** — EU AI Act risk tiers and Annex III high-risk categories; GDPR Arts. 15, 22, 33 and 35
- **The AI lifecycle** — development, deployment and end-user serving

Topic 2 — Explainable AI (XAI)

- **Interpretable vs. black-box models**
 - **Post-hoc explanation methods** — LIME, SHAP, partial-dependence plots, counterfactuals
 - **Intrinsic interpretability** — decision trees, rule lists, monotonic and generalised additive models
 - **Global vs. local explanations** — whole-model vs. single-decision
 - **What makes a good explanation** — fidelity, stability, human understandability
 - **Audience-specific explanations** — the affected person, the operator and the regulator
 - **Explanation misuse** — attributions explain the model, not the world; a plausible explanation can increase over-trust
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Topic 3 — Causal AI for Deeper Trust (*optional*)

- **Correlation vs. causation**
- **The ladder of causation** — association -> intervention -> counterfactual
- **Structural Causal Models (SCMs) and DAGs**
- **Reading a graph** — confounding, colliders and mediation
- **Identification** — the estimand vs. the estimate; ITT vs. TOT
- **Causal failure modes** — confounding as the default explanation for observational association
- **Why it matters** — causal explanations are more reliable and actionable

Topic 4 — Engineering Track: Prompt & Agentic AI Safety

- Preventing **prompt injection**
- **Vulnerability identification** in LLM applications
- **Data privacy** and governance — what to provide and what not to
- **PII** — what it is and how to handle it
- Handling **sensitive information**
- **Input/output validation** for AI and LLMs
- **Bias detection**
- **Alignment** (see Anthropic's alignment work)
- Handling **safe-request refusal**
- **Hallucinations** and how to detect them
- **Alignment and autonomy control** — human in / on / out of the loop
- **Governance**
- **Data poisoning**
- **Incident management** — who bears the cost when losses occur

Topic 5 — Integration, Auditing & Regulation

- **Accountability as a mechanism** — a named human-oversight role with authority to override

- **The evidence pack** — model card, DPIA, technical documentation, fundamental-rights assessment, post-market monitoring and incident reports, right to explanation
- **Post-market monitoring** — every signal has a threshold, an owner and an action
- **Incident response and the incident loop** — severity, a clock, and feeding fixes back into the test suite
- **Go / no-go and residual-risk acceptance** — escalating risk tolerance to a human for signature
- **Decommissioning triggers** — decided before a crisis, not during one
- **Red-teaming** — adversarial testing of the system

Topic 6 — Testing & Best Practices

- **Test strategy for AI systems** — unit, property-based, integration and load testing
 - **Monitoring & observability** — what to log, measure and alert on
 - **Deployment practices** for trustworthy systems
 - **Architecture decision records (ADRs)**
 - **The best-practice checklist** — prompt engineering, data privacy & RAG, agentic safety, testing, monitoring, deployment, ADRs
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Resources + Additional Materials

References & further reading

- **NIST AI Risk Management Framework (AI RMF 1.0)** — GOVERN / MAP / MEASURE / MANAGE.
- **EU AI Act** — Annex III high-risk categories, Chapter III obligations, Arts. 27, 72–73 and 86.
- **GDPR** — Arts. 15, 22, 33 and 35.
- Christoph Molnar — *Interpretable Machine Learning*.
- *Causal Inference for the Brave and True* (Matheus Facure) · DoWhy / EconML tutorials.
- **OWASP Top 10 for LLM Applications** — prompt injection, excessive agency, sensitive-information disclosure.
- Anthropic alignment material — RLHF, RLAI, Constitutional AI, Responsible Scaling.
- *Trustworthy AI* (module reference book).

Additional materials

- Module plan — *Responsible and Trustworthy AI*.
- Best-practice checklist — the seven-area "never / always" checklist.
- Question banks — core questions (C1–C4), session questions, and the agentic and scenario question sets.